

Lost & Found Tracker

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Lost & Found Tracker

Introduction

Lost & Found Tracker is a mobile application that allows users to report, browse, and track lost or found items within a community. Built using Android Studio and integrated with Firebase services including Authentication, Realtime Database, and Storage, the app promotes transparency and collaboration by connecting individuals who lose belongings with those who find them.

Purpose

The application provides a centralized platform for managing and tracking lost or found items. It is beneficial for places like universities and offices where item loss is frequent. Users can securely register, report item information, and view reports shared by others.

Core Features

- User Registration and Login via Email & Password
- Secure Firebase Authentication
- Report Lost or Found Items with Title, Description, and Image
- Upload Item Images (Camera or Gallery)
- View All Reported Items in Real Time
- View Detailed Item Info on Tap
- "My Items" Section to Track Personal Submissions
- Firebase Realtime Database for Instant Data Sync
- Firebase Cloud Storage for Storing Item Images
- Material Design UI for Clean User Experience

Technology Stack

Component	Technology Used
Frontend	Android (Java, XML Layout)
Backend/Database	Firebase Realtime Database
Authentication	Firebase Authentication
Media Storage	Firebase Storage
IDE	Android Studio
UI Design	Material Design Components

Lost & Found Tracker

App Workflow

Authentication Flow

The application begins with an authentication process where users are required to either log in or register. Upon registration, a user account is created using Firebase Authentication. Once logged in, users gain access to all app functionalities including item reporting and browsing. The login system is secured using Firebase's email/password system, ensuring that data is accessible only to authenticated users.

Item Reporting Flow

Users can report items that they have lost or found. The reporting screen allows them to enter item details such as the name, description, and location of the item. They can also upload an image from the camera or gallery. Submitted items are stored in Firebase Realtime Database, and associated images are uploaded to Firebase Storage. Each item is linked with the user's unique ID to support future retrieval.

Item Viewing Flow

The app provides a live feed of all reported items which are updated in real time using Firebase's Realtime Database. Users can scroll through the list to view items reported by others. Tapping on any item reveals the full details, including the image, description, and contact information if applicable. A separate section also exists to show items reported by the currently logged-in user.

Firebase Integration Details

Firebase Authentication

This service handles all user login, registration, and session management. It secures the application by ensuring that only authenticated users can report or view items. Passwords are never stored locally and authentication is managed securely by Firebase.

Firebase Realtime Database

All item details submitted by users are stored in a cloud-hosted NoSQL database that updates in real time. Each item entry includes a title, description, location, image URL, timestamp, and user ID. The data is structured in a way that supports efficient searching and filtering in future enhancements.

Firebase Storage

User-uploaded images are stored in Firebase Cloud Storage. Each image is associated with a unique URL, which is then linked with the item entry in the database. This setup ensures efficient media handling and optimized retrieval.

Output Screen:

Lost & Found Tracker

Email

Password

LOGIN

Don't have an account? [Register now](#)

Email

Password

Confirm Password

Register

[Forgot Password?](#)

Already have an account? [Login here](#)

HomeScreen:

9:19 AM

LOST & FOUND TRACKER

+

Report Lost Item

🔍

Browse Found Items

📁

My Reported Items

LOGOUT

pen
blue color
Location: classroom
Posted: Jul 10, 2025 08:16 AM

camera
in black color
Location: Islamabad

9:21 AM

laptop

grey color

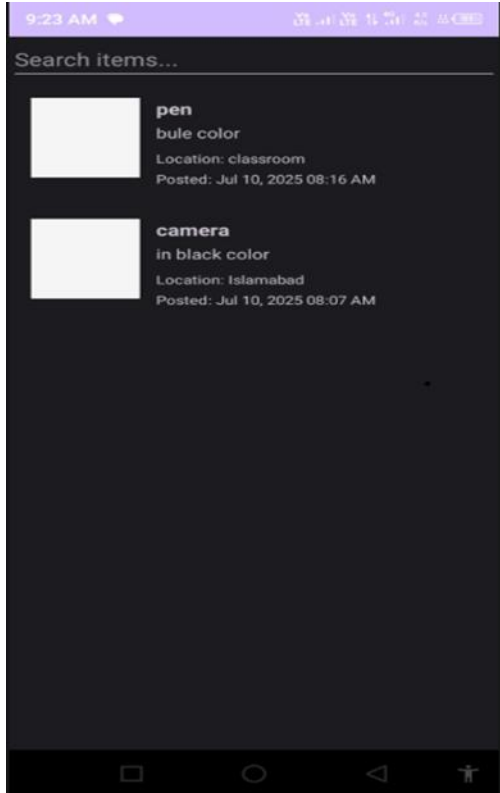
library

Upload Image

Submit Report

Lost & Found Tracker

Browse lost item:



Future Improvements

- Add real-time notifications for item matches
- Implement search and category filters
- Integrate map and GPS support for location-based search
- Add admin panel for item moderation
- Introduce chat feature for user-to-user communication
- Enable archiving or auto-deletion of old reports
- Implement biometric login for added security

Conclusion

Lost & Found Tracker is a robust, community-centric Android application designed to bridge the gap between individuals who have lost personal belongings and those who have found them. With a cloud-based backend, user authentication, and real-time updates, the application ensures security, scalability, and ease of use. Its design and implementation make it ideal for universities, office campuses, and public institutions. The project demonstrates effective use of Firebase and Java, adhering to Android development best practices and providing a foundation for future enhancements.